The Vessel

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The chapters before this one built a world from two moves and priced it. This chapter stops building and starts watching. We will pour thousands of terms into a sealed vessel, drive it, and see what kind of order forms on its own, with nobody in charge. The word for this is self-organization: order that arises without an orderer.

One promise before the vessel is opened. To describe what happens inside it, this chapter will find itself reaching, again and again, for the vocabulary of chemistry: atom, molecule, bond, reaction, temperature, phase. These are loans, and the book's contract against smuggled vocabulary applies. So the chapter keeps its ledger of words: every chemical word is marked at its first use with its exact formal meaning, and at the chapter's last door the loans are counted. But this chapter's ledger has a destination the last audit could not have had. When the counting is done, we will find that every single loan fit, strangely well, and the chapter will end on the question that fit forces: whether the road we are walking, the road of pure form under a conservation law, could be walked further, all the way to an understanding of real atoms and real molecules. That question is the summit of this chapter. The pages before it are the climb.

The Question of the Atom

Chemistry began as a claim about combination. Democritus guessed it in antiquity: beneath the endless variety of stuff, a small number of indivisible units, and all difference a difference of arrangement. For two thousand years this remained philosophy. Then, in 1808, John Dalton made it bookkeeping. Elements combine in fixed ratios, he showed, always the same small whole numbers of this per that, exactly as if matter came in unit pieces that clip together in definite ways. Water is two of one and one of the other, every time, everywhere. The molecule entered science not as a picture but as an arithmetic fact.

And then something happened that belongs in this book's ancestry. The mathematician Arthur Cayley had planted a quiet idea back in 1857, a family of branching figures he named trees. In the 1870s chemistry came knocking with a counting problem: how many distinct molecules can a given handful of carbon atoms form? Cayley answered by growing his trees into an enumeration method, counting the alkane isomers branch by branch, and tree counting has been mathematics ever since. The reader has met trees; the numbers of this book live in them. Their arithmetic came of age counting molecules. Chemistry, pressed on the question how many kinds, answered with combinatorics, because at the level where the question lives, chemistry is combinatorics: fixed kinds of unit, fixed rules of joining, and the whole zoo of substances a census of arrangements.

Yet through all of this, into the twentieth century, serious scientists refused the atom. Ernst Mach's challenge was famous and fair: have you seen one? An accounting trick, he held, is not an ontology. The matter was settled the hard way, and a life was spent on the way to the settling: Ludwig Boltzmann, who had built the statistics of arrangement that this whole chapter runs on, died in 1906, worn down in part by decades of that very skepticism, one year before the proof arrived. In 1905 Einstein showed that if atoms are real, the trembling of pollen grains in water, watched for eighty years without explanation, must have an exact statistical signature; in 1908 Jean Perrin measured the trembling, extracted the count of atoms per mole, and got the same number by a dozen independent routes. The units are real. What had won, though, is worth saying precisely, because it is the hinge of this chapter: what won was not a substance but a scheme. Discrete units of few kinds;

combination under fixed local rules; large-scale order as the statistics of arrangement. Strip the scheme of all its physics and ask: how much of chemistry's furniture appears in any world built like that, even a world of pure form with no matter in it at all? We own such a world. It has exactly two kinds of unit and two rules of transformation. Pour it into a vessel and look.

Notes and sources. Dalton's law of multiple proportions, 1808, in the New System of Chemical Philosophy. Cayley introduced trees in 1857 and applied their enumeration to the isomer problem in the 1870s (On the mathematical theory of isomers, 1874, and the alkane counts of 1875); the role of chemistry in maturing tree enumeration is standard history of combinatorics. Boltzmann's death in 1906 and its timing before Perrin's vindication are standard history, reported here without a causal claim beyond the sources' own. Mach's skepticism, Einstein's 1905 Brownian-motion paper, and Perrin's 1908 measurements, gathered in his book Les Atomes (1913), are standard history of physics. The scheme-not-substance reading of the atomists' victory is this book's phrasing.

The First Evening

Here is the first vessel, and its rules are deliberately loose. The ledger opens. Call the two letters of the calculus atoms: the formal cash value is exact, an atom is one leaf, S or K, and there are no other kinds. Call a closed term a molecule: a finite build of atoms joined by application. Six hundred molecules float in the vessel. Time is a sequence of collisions: two molecules meet, one is applied to the other, the result is computed and joins the population while a random molecule is washed out, so the vessel stays at six hundred. And in this loose vessel, copying is free: if a computation calls for a second copy of a piece, the copy simply appears; what is discarded vanishes without trace.

We filled this vessel three times from three random starts, and the same drama ran three times. Among the molecules are pass-through species, terms with the property that whatever meets them, the collision's product is a copy of the partner; they convert each encounter into more of the other. For a short opening stretch these transmitters flourished, and variety flourished around them. Then one compact species took the vessel. The papers call it the absorber, tau, and the name is a description: a molecule whose every collision-product is itself. Whatever it meets, the result of the meeting is one more tau, and nothing of the partner appears in it. Its four-step reaction cycle is a compiled theorem in the program's module, not an anecdote. In the loose vessel this species is unbeatable: in all three runs its population grew along the same relentless curve until it held all six hundred places. Total monoculture. A vessel of free copying always ends as a crystal of its best converter, and the transmitters are the first species it consumes.

Now hold the story and ask the accountant's question, because the whole chapter turns on it. Every collision made a new tau, and a new tau is atoms joined together. Who joined them? Where, physically speaking, did the copies come from? In the loose vessel the answer is: from nowhere. The rules granted them free. The absorber did not win by depth; it won because this vessel handed it a money printer, one economics loan, cash value: a rule that grants copies at zero cost. No real chemistry has such a rule. In real chemistry, matter is conserved: every molecule on the right of a reaction is built from atoms that stood on the left. So the real experiment is obvious. Impose the chemist's rule, and see what any species is worth when every atom must be accounted for.

Notes and sources. The flow reactor, the three takeovers, and the two absorber species are the working paper Combinator Soup (v0.3); the compiled anchors are tauAbsorbs, tauFix, and tauNormal. The selection mechanism in one sentence, the transmitter donates its collisions, the absorber keeps them, unconstrained selection finds the keeper, is that paper's.

The Law of the Mint

The honest vessel runs on the chemist's rule: nothing enters, nothing leaves, every atom accounted. Six thousand atoms, three thousand of each kind, and molecules now condense from the pool and cleave back into it. To say what this does to the possibilities, the ledger takes one more loan, the load-bearing one. Call each application joint in a molecule a bond, and picture it with the toy box: a molecule is a build of interlocking bricks, a bond is one click between two bricks, a lone atom has no clicks, and a build of ten atoms has nine. The count of bonds is the count of assembly acts that built the thing. Whoever can create bonds can create structure, and whoever can create structure can, in principle, create copies of anything.

The honest vessel has four kinds of event, and each has a fixed effect on the bond count, checked by machine, true in every run the vessel could ever have. Condensation, the background event that presses two molecules into one, creates exactly one bond. Cleavage, its opposite, destroys exactly one. The dropping move, our K, working inside a molecule, destroys exactly two bonds and creates none, releasing the dropped part whole into the vessel. And the sharing move, our S, the only copier in the whole world, is not allowed to conjure its copy: it must find a spare copy of the needed part floating free in the vessel and bind it, and when the accounting is done, the copy it installs carries exactly the bonds the recruited spare carried. Not one more. These four statements are the vessel's stoichiometry, and the word is meant literally: each is a balanced equation, left side weighing exactly the right side plus what is released, compiled in the module the way every claim in this book is compiled. The sharing move does not manufacture structure. It moves structure.

Add the four lines and you have the law, and the law is the chapter's center. In the honest vessel, no reaction ever creates a bond. Not one, not ever, by any pathway. A molecule can react for a thousand steps, rearrange, split, recruit, and at the end of all its steps it will not have minted a single bond that did not already exist. The only bond-maker in the world is the blind background condensation, the driving of the vessel, which favors no species and takes no orders. Call that drive the weather, cash value: the undirected stream of condensation and cleavage events. In the program's module the law is one theorem quantified over every term and every possible sequence of moves: along any computation whatsoever, the minting meter reads zero. Building belongs to the weather. Reaction can only spend.

And now look back at the loose vessel with the law in hand, because one line of accounting explains everything we saw there. Write the sharing move's balanced equation, then delete one item from the left side, the recruited spare. What remains is a sharing move whose copy costs nothing, whose bonds appear from the void. That is the loose vessel, exactly. One equation, two worlds: with the spare, sharing moves structure; without it, sharing mints it. The absorber's monoculture was never chemistry. It was what any species looks like while it is holding the mint.

Notes and sources. The four balance equations and the never-minting theorem are compiled: condBonds, kBonds, sBonds, and stepsNeverMint, in the working paper The Assembly Ledger (v0.1). The strong reading of the sharing balance, the installed copy's bonds equal the recruited spare's bonds exactly, is the sBonds equation itself; the one-equation-two-worlds reading is that paper's Section 2. Bonds are application nodes; atoms are leaves.

The Furnace

So run the absorber through the honest books. Its four-step cycle, one share, two drops, one share, goes under the meters. Minted: zero, as the law demands of every reaction. Burned: four bonds, two per drop, gone from the vessel's stock. And the yield all this burning buys: one tau at the end, from one tau at the beginning. Net new copies: none.

Read the verdict slowly, because it is a deduction, not an observation. In the honest vessel the absorber cannot multiply, and the reason is not that we watched it fail; it is that multiplying would require minted bonds, and no reaction mints. Each cycle it destroys four bonds of the common stock and produces no second copy: an exothermic species, all release and no assembly. The ledger takes the loan furnace for it, cash value: a species whose reaction cycle burns bonds and mints none. And the measured history agrees with the deduction completely: across every long run of the honest vessel, at every checkpoint, the tau count is zero. The monoculture of the loose vessel never even begins.

What fills the vessel instead, the measurements say, is a standing population of small, reactive molecules, and at its heart, five persisting copies of a three-atom species the reader knows: S K K, the crown, identity itself, the purest of the pass-through terms and the first species the loose vessel consumed. The law explains the fit without any appeal to virtue. The dropping move releases what it discards, so every drop restocks the vessel's pool of free parts, and the pool is what condensation and recruitment both draw on. A species persists here not by what it keeps but by what the network can rebuild: the standing population is the set of molecules the vessel's own reaction traffic continually re-makes from the common pool, a steady state in the chemist's exact sense. Persistence belongs to remakability. The absorber, which only burned the pool, was never remakable from it, and the vessel forgot it without ever having met it.

Notes and sources. The metered cycle is compiled, tauMealMint and tauMealBurn, in The Assembly Ledger, Section 4. The extinction data and the standing population, identity at population five in a vessel of roughly five hundred seventy, are Combinator Soup (v0.3), Sections 6 to 8, illustrations under the law; that paper's own caution about small-species counts stands, and the steady-state reading is stated there as the conservative one.

Glass, Gas, and the Busy Middle

The law says what can never happen. To see what actually does happen, we watch one particular honest vessel, and everything in this section is measurement, local to this vessel and its settings, and labeled as such.

Left entirely alone, the honest vessel does not stay busy. It locks. Molecules condense, grow, bind the material into large one-of-a-kind lumps, and the reaction count falls to zero: structure without change. The vessel must be driven, kept stirred by ongoing condensation and cleavage, or it stops. And the vessel taught us the sharper fact: being driven is not enough. It matters how the drive is balanced.

Give the vessel a dial, and take the chapter's next chemistry loan to name it: temperature, cash value, the melting share, the fraction of drive events that cleave rather than condense. At zero, everything only condenses: the locked state again, two hundred fifty six molecules, every one unique, and in the last measuring window nearly fifteen thousand sharing reactions attempted and every single one failed, because a vessel of large unique lumps holds no free spares to recruit. Plenty of reaction sites, no parts. Turn the dial high and everything only shreds: four and a half thousand fragments averaging little more than one atom, variety collapsed, almost nothing even attempting to react, because fragments have no reaction sites. Plenty of parts, no sites. Two opposite ways for chemistry to stop.

The reader has seen these two states through a window, and the ledger should say so plainly: they are the solid and the gas. A solid is structure that cannot rearrange; a gas is mobility that cannot assemble; and every chemist knows where reactions actually live: in between, in solution, in the melt, in the liquid regimes where parts are both bound enough to matter and free enough to move. Our vessel, which contains no matter at all, reproduces the arrangement:

melting share	activity	kinds of molecule
0	0	256
1/5	1,094	322
1/3	1,629	357
1/2	1,720	395
2/3	805	238
4/5	238	123

Activity, the count of reactions firing in the final window, rises, peaks, and falls, and the peak sits at the balanced share, one half, where taking-apart matches putting-together. The count of distinct species peaks at exactly the same setting, and a second, independently seeded run agrees on both peaks. The mechanism is two plain curves multiplied: hotter vessels attempt fewer reactions, because fragments offer no sites, while each attempt succeeds more often, because melting fills the pool with spares. Attempts fall, success rises, attempts times success peaks in the middle. The vessel is busiest exactly where it is most various, starved of parts on one side and of sites on the other. A phase structure, with a reactive middle, in a world with no matter in it: the loan word phase is hereby marked, and the reader is asked to hold the strangeness of the fit until the last section, where it becomes the point.

Notes and sources. The sweep, both seeds, the anchors, and the attempts-times-success decomposition are the working paper Freezing and Flux (v0.1); the table is seed eleven, final window, with seed seventy seven peaking at the same setting. Measurements of this vessel under these rules, sitting here as the law's illustration, not as its evidence. The solid, liquid, gas reading is the ledger's loan, marked here.

What Rose, and the Question

Count the harvest, and count it as an audit of the ledger, loan by loan. Atoms: borrowed, and the vessel's version is two kinds of leaf under an exact conservation law, verified to the last unit at every checkpoint. Molecules: borrowed, and the vessel's version is finite builds with a census. Bonds: borrowed, and the vessel's version is assembly joints under balanced equations, stoichiometry compiled. Reactions: borrowed, and the vessel's version comes with a law real chemistry also obeys, that reactions spend structure and never mint it, with the absorber's extinction as a corollary and every self-copying structure anyone's vessel has ever grown, including the self-reproducing metabolisms reported in Kruszewski and Mikolov's neighboring system, paying for every copy bond by bond out of what it takes in. Temperature and phase: borrowed, and the vessel's version is a dial with a solid at one end, a gas at the other, and a narrow busy middle where reaction and variety peak together. Steady state: borrowed, and the vessel's version is the standing population that persistence-by-remakability maintains.

Every loan fit. That is the fact the audit leaves on the table, and the chapter owes the reader both readings of it. The deflationary reading: of course the loans fit, we chose them, and an author with a motive can dress any system in any vocabulary. The ledger was kept precisely to give this reading its full force, and nothing in the measurements refutes it. But there is a second reading, and it is the question this chapter was built to reach. The scheme that won the atomic debate, discrete units of few kinds, combination under fixed local rules, large-scale order as statistics of arrangement, is not something we projected onto the vessel. It is what the vessel is. And chemistry's own history keeps confessing that at its structural level it is made of exactly this scheme: Dalton's ratios are counting; Cayley's molecules were counted with trees; the periodic table is a classification of combining be-

havior; a balanced reaction is bookkeeping over units. So the question, stated with the care it deserves: how much of real chemistry is combinatorial theorem in disguise? Which of its laws would hold in every possible world of discrete combination under conservation, and which need the specific physics of our matter, its quantum mechanics, its geometry, its three dimensions? Take one concrete unknown as the measure of the gap: valence, the fact that each kind of real atom accepts just so many neighbors and no more, carbon four, oxygen two, is exactly the kind of law our vessel does not yet exhibit and would most need to. The vessel gives the question standing, because it exhibits, in a world of pure form, conservation, stoichiometry, an extinction law, a steady state, and a phase structure, none of them put in by hand. It does not answer the question, and this book will not pretend it does. The honest boundary is easy to state: our bonds have no energies, our atoms no orbitals, our vessel no space, and the periodic table's actual content, why these kinds and these valences, lives exactly there, on the far side of what combination alone has yet reached.

So the chapter ends by registering a destination rather than claiming an arrival. The program's research agenda now carries the question as an open line: to take the road further, from this vessel toward real atoms and molecules, and to find which chemical laws are theorems of combination itself, provable in the module like everything else here, and which are matter's own signature. If that mathematics succeeds, a later chapter will report it, under this book's usual contract; if it fails, the failure will be reported too, and the boundary will have been drawn honestly. Until then, one reading is offered, once, and the reader's chair decides its fate: the vocabulary of chemistry fit this vessel because form is not imitating matter here; it may be the layer matter's own order is written in. Structure first, substance as its wearer. The traditions that this book keeps company with have always said something of the kind. The vessel neither proves them right nor needs them to be; it only makes the question askable with theorems in hand. And one resident is still missing from the vessel, with a stated price on its absence: nothing here carries its own description and builds itself from it, and by the law of the mint, anything that ever does will do it the honest way, bond by paid bond. Building exactly that machine, and proving it does what it claims, is the next chapter: the machine that runs itself.

Notes and sources. The circulation results and the controlled comparison are Combinator Soup (v0.3), Sections 9 and 10. Kruszewski and Mikolov, Combinatory Chemistry (ALIFE 2020) and Emergence of Self-Reproducing Metabolisms (Artificial Life 27, 2022), whose statement that the S reduction is the sole incorporation channel is paraphrased; the universal bond bill is The Assembly Ledger (v0.1), Section 5. The chemist's-question research line is registered on the program's agenda as of this draft, first target to be scoped there, not here. The final reading is governed by the book's front note; the deflationary reading is printed beside it on purpose.